The suggested Bi-Slant Conveyor Belt Toilet, eliminates the need of flush water apart from providing the separating technique for urine from feces. Conveyer belt carries and delivers feces into the back tank, while urine slopes into front urine tank. By adding micro organism bio seeds into tanks, they can be made bio tanks utilized as fertilizers.

(c) Solar-Powered, Self-Cleaning Toilet.

Sleek and made of stainless steel, these toilets are designed to be installed in places where access to electricity and common sanitation methods is difficult, if not impossible. What makes these toilets even more user-friendly is that they can be located through an Android app called e-Toilet. Spread over 35 square feet, each automatic unmanned toilet cleans itself before entry and after use.

They come equipped with sensors for light and exhaust fan to ensure that they consume less power and water than conventional toilets. Before and after every use the toilet flushes automatically, which requires minimum amount of water. The amount of water required is computed from samples obtained from sensor. Every flush on an average uses 1.5 litres of water, compared to the 8-10 litres used by a ordinary flush. Its floor is washed automatically after every tenth use. The lights turn on automatically and draw power from a built-in solar panel.

Also, there are provisions for waste treatment using anaerobic bio-degradation. These e-Toilets have the added advantage of being eco-friendly, with provisions for waste treatment using anaerobic biodegradation.

(d) Ensuring sanitation of toilets using IoT.

In large premises like hospitals, railway stations and educational institutions many a number of toilets are there whose sanitation supervision is difficult by physically visiting them. The difficulty can be reduced by placing odor sensors in toilets, getting sensors equipped with communication modules connected to WLAN to get the notifications in the institute monitoring server/system when odor intensifies.

4.7 SEWERAGE SYSTEM

Wastewater generation scenario in Indian city and its treatment

In India, the estimated sewage generation from Class I cities and Class II towns (representing 72% of urban population) is 38,524 million litres/day (MLD), of which there exists treatment capacity of only 11,787 MLD (about 30%).The 35 major metropolitan areas (with population over 1 million) have a collective sewage treatment capacity of just over 50%, but high variability exists among them. Only five metro cities have treatment capacity close to 100% of their sewage generation, these are Hyderabad, Vadodara, Chennai, Ludhiana and Ahmedabad. Delhi has the largest sewage treatment capacity in absolute terms, but it is only about 60% its needs, while Mumbai has the second largest treatment capacity in absolute terms but meets only 80% of its needs. Almost all other major metropolitan cities have treatment capacities below 50% of their sewage generation.

There have two type of challenges in wastewater treatment trade.

(a) Onsite challenges

(b) Offsite challenges

(a) On site challenges

The conventional engineered wastewater treatment system is extremely expensive and requires complex operations and maintenance cost of expanding sewer networks, which are very rudimentary or non-existent in many Indian cities. That is mostly started off as unplanned settlements. The total capital cost of establishing collection and treatment systems for the entire urban wastewater generated is much more than what the government plans to spend; as a result progress in increasing coverage is likely to be slow in the foreseeable future. Like any other infrastructure project in India, land conflicts/disputes often hold up sewage treatment plan(STP) construction long after funds have been sanctioned. STPs run below installed capacity is the lack of adequate sewer connection, frequent power cuts, lack of skilled personnel, inability of urban local bodies (ULBs) to afford operation and maintenance costs.

(b) Offsite challenges

Analysis exhibits that macro factors such as FTA (Free Trade Agreement), cost sensitivity, and varied climatic and usage considerations directly impact operations in India. There exists immense opportunities for foreign companies owing to:

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- Commercially viable advanced technology
- Environmentally conforming solutions

However, foreign companies need to overcome concern such as :

- Lack of Indian experience to meet tender requirements
- Different waste treatment procedure

✓ **Need For Wastewater Treatment**

Waste Water contains a variety of organic and inorganic substances dissolved which also includes toxic elements such as Cadmium(Cd), Zinc(Zn), Copper(Cu), Chromium(Cr), Ar-senic(As), Mercury(Hg), Lead(Pb) etc. These may not be in present in dense concentration to affect human life but may affect plants and animal life. Disposal of the waste water directly into water bodies like rivers and oceans may affect the aquatic animal life. The water in these water bodies are consumed by animals and also by the people living downstream. Consumption of this contaminated water leads to health problems in a long run. The water in a river or lake id normally utilized by farmers for cultivation and the presence of Micro and Macro Organisms in the water affect the agricultural growth.

✓ **Sewer Network – The Present Indian Scenario**

The proper sewerage system includes collection of sewage from source of generation through sewer network, treatment using appropriate available economical technology to the prescribed disposal standards and safe disposal to natural water bodies.

There are two systems available in general for Waste Water Collection and Disposal, viz.

(i) **Separate System** – sewage and storm water are collected separately & sewage is treated for safe disposal

(ii) **Combined System** – sewage and storm water are collected together in same drain & treated together. Presently the separate system is preferred due to its technical and economical advantages. Hence, in India also, nowadays almost all the sewer networks are designed for separate systems. Almost 80% of India's population does not have proper sewer systems1 . As many as 55% of rural households defecate in the open and about 12.6% of urban households defecate in the open. This number is higher for slums, with 18.9% of households defecating in the open2 . Even under the Swatch Bharath Mission(SBM) also it is observed that as per the 2011 census around 8 million households of the 4,041 statutory towns in India have no access to toilets and continue to practice open defecation. Sanitation facilities are almost absent in rural areas, where around 69% of India's population resides.

The following are the ***major disadvantages*** or challenges being faced due to lack of proper sanitation facilities.

i. Population affected with health issues and ultimately affecting GDP of the country
ii. The ground and surface water sources are being polluted and hence water supply is becoming costly & inadequate

The major ***reasons for inadequate sewer networks*** / sanitation facilities are as follows.

i. Insufficient funds
ii. Improper design approaches & management of sewer network systems

✓ **Measures for economical sewer network:**

(a) Selection of appropriate pipe material and usage

The following are the various parameters need to be considered while selecting pipe material for sewer networks.

- Load bearing capacity of the particular class of pipes
- Various loads transferred on the pipes
- Least life cycle cost (LCC) of pipe
- Internal pressure to be withstood
- Requirement of proper bedding

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- Frictional losses or roughness coefficient etc.
- Type of soil and it's characteristics, presence of any chemicals.

Based on the above considerations, the required type of pipe shall be selected out of the commonly available pipes – Vitrified Clay Pipes (VCP), Reinforced Cement Concrete (RCC) pipes, High Density Poly Ethylene (HDPE) pipes, Double Wall Corrugated (DWC)- HDPE, Ductile Iron (DI) Pipes, Mild Steel (MS) pipes etc. At present, although VCP pipes up to 380 mm dia. are cheaper and suitable for gravity sewer application up to certain depths.

(b) Spacing of Manholes

In any sewer network system, manholes are a costly item and it is advantageous, cost wise, to reduce the number of manholes to the extent possible. As per Central Public Health and Environmental Engineering Organization (CPHEEO) (2013), provision of manhole spacing along the street sewers (which are directly connected to households) must not be more than 30 meters.

(c) Optimization of Intermediate Pumping Stations (IPS)

Based on the topography of catchment of network, the necessity of either IPSs or lifting stations generally arises for reducing the overall depth of sewers or meeting the restricted depth criteria due to high ground water table etc.

Atal Mission for Rejuvenation and Urban Transformation (AMRUT)

Five hundred cities will be taken up under AMRUT. Under the sewerage component of AMRUT, the goals are:

(i) Decentralized, networked underground sewerage systems, including augmentation of existing sewerage systems and sewage treatment plants.

(ii) Rehabilitation of old sewerage system and treatment plants.

(iii) Recycling of water for beneficial purposes and reuse of wastewater.

Smart wastewater management is defined as the use of technology to optimize sanitary network performance and wastewater operations to supply reliable sanitation for communities and pollution-free clean water. Smart wastewater systems perform by detecting and controlling combined sewage overflows and chemicals in wastewater with the help of IoT sensors to meet the demand for freshwater in smart cities. The major utilities of smart wastewater management are as follows:

- Remotely monitor and detect problems to prioritize and recognize maintenance issues.
- Consumers can be provided with tools and information to enable them to make informed choices about their behavior patterns.
- Comply with policies and regulations on water quality and conservation transparently and confidentially.

✓ **IoT based smart wastewater management systems:** **(Dec.-2021, June-2022)**

Smart wastewater systems can meet the demand for freshwater in smart cities by detecting and preventing combined sewage overflows and chemicals in wastewater with the help of IoT sensors.

Reliable communication between wastewater management systems and administrators brings utility and cost savings together. It helps to find areas where wastage of water is likely and take quick actions to minimize water wastage.

System Architecture: In this diagram the various sensors such as ph sensor, salinity sensor, gas sensor, water level sensor are connected to the PIC controller, the internet of things (IOT) will detect the waste water condition through sensors whether the condition is normal or abnormal by using sensor ranges and the information will get stored into cloud from the cloud the information can be accessed by multiple users through web service and mobile application.

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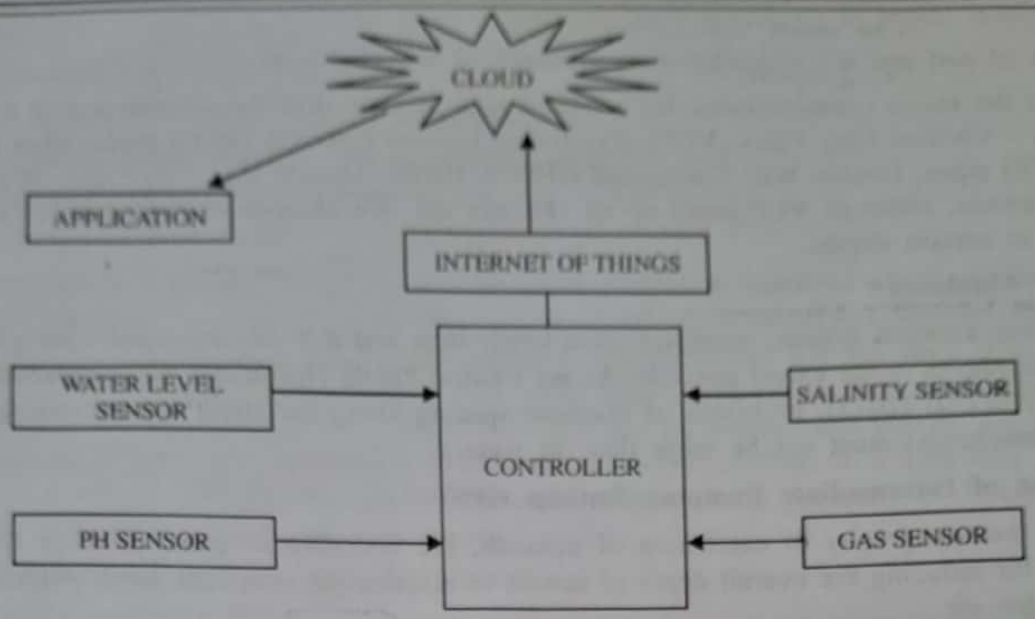

Fig. 4.6 System architecture for IoT

Some of the systems are:

1. **Preventing CSOs (combined sewage overflows)**

The IoT sensors are capable to detect combined sewage overflows. You can set threshold values with the help of smart solutions and get a real-time notification as soon as the sewage water overflows. This helps you preserve combined sewage overflows and reduces water wastage at a significant level.

During rains, rainwater results in an overflow of combined sewer systems in poor wastewater management systems. Monitoring the inflow of water with the help of IoT sensors can help water utilities to detect overflows in real-time and take appropriate actions to reduce water wastage. For instance, administrators can control valves and provide more water flow to areas for a specific time interval to prevent CSOs. Administrators can also automate the control of valves with the help of AI technology so that the valves adjust themselves according to the inflow of water. And also, ML algorithms can help predict the inflow of water and prevent CSOs.

2. **Tracing chemical detection**

IoT can play a significant role in detecting chemicals in water pipelines. With non-IoT wastewater management systems, researchers would have to detect chemical levels in water manually, which can be a time-consuming process. IoT sensors can not only easily detect the presence of chemicals, but also send alerts to a remote dashboard. Detecting traces of chemicals early will help to optimize the wastewater treatment processes and ensure the release of chemicals for treatment is within limits.

3. **Replacing leaky pipes**

A leak in distribution pipes that distributes water regionally can result in heavy loss of water. Hence, these pipes need to be changed regularly according to necessity. IoT sensors can collect data like type of sand that pipes rest on, topography, and weather records. And then AI can help analyze these data to find patterns that offer clues to the water utilities about which pipes are at risk of leakage.

4. **Select best treatment process**

Specific constituents require particular treatment processes to be removed. The advanced analytics feature of an IoT solution can recommend a series of apt processes based on the concentration of water. This lets the treatment plant use minimum processes to remove excess amounts of contaminants from the wastewater. Processes in the treatment plants are optimized and operation costs are reduced.

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✓ **Advantages of using smart wastewater system :**

(1) Real-Time Data Integration:

Real-Time Data can be transformed into valuable information for quick decision making in the areas of utility outside the control room. The utility team can stay aware of what is going on in the field every time and can respond immediately and appropriately to any problem by migrating the real-time information out of the control room. Whenever there is a pipe leakage or a faulty piece of equipment, the utility team can dispatch crews immediately to the spot with the specific information required.

(2) Enhancing Customer Satisfaction:

Having a flexible system in place allows utilities to set their future goals. They can gain a more precise view of production versus consumption by importing data from the Customer Information System (CIS) into the SCADA system. Utilities can even communicate with customers accurately by linking hydraulic model simulations with the CIS.

(3) Accurate Forecasting of Demand:

Utilities depend on historical data to create a demand curve so that they can properly adjust their production to make sure they are operating at peak efficiency. However, the process can be automated by using an AMI system so pumping regimens can be adopted to more appropriately fit the demand. A Demand Management System (DMS), on the other hand, can track the water demand where utilities can program pumps to cater to the specific needs of saving water and energy.

4.8 FLOOD MANAGEMENT :

India is highly vulnerable, as most of its geographical area is prone to annual flooding. The high losses and damages due to floods show the poor adaptation and mitigation status of India and inadequacy in disaster management and preparedness. Thus, there is a need for an Integrated flood management system.

(a) Disaster Preparedness Plan : A comprehensive flood management plan is needed to include Disaster preparedness. This may require strengthening of the following:

- Flood Hotspot Mapping at local and regional scale.
- Management and regulation of riparian zones to prevent spilling and erosion.
- River flood modelling to prepare for incidences like reservoir breach and emergency water release from dams.
- Advanced techniques such as mapping based on satellite imagery and **Geographic Information Systems** will help in development of flood early warning systems.

(b) Integrated Approach : Steps need to be taken for watershed management through an integrated approach. Often these approaches involve both hard engineering solutions and ecologically sustainable soft solutions.

- **Hard Solutions:** It involves civil engineering construction such as dams, culverts and dykes, widening and deepening of river channels and diversion channels to store and divert water to increase the lag time of water reaching downstream.
- **Ecological Soft Solutions**: The solutions such as restoration and management of riparian zones, Afforestation along the river channels which led to retention of rainwater and reduces the river discharge.
- An integrated approach to managing floods requires a sound understanding of the patterns that rivers such as the Ganga and its tributaries display during the monsoon.

(c) Prioritizing Buffers, Flexibility and Adaptability : This includes reviewing safety criteria of dams and canals, re-building these with higher safety factors, creating new intermediate storages, and introducing dynamic reservoir management.

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(d) Reducing Disaster Risk Reduction : There is a need for efficient implementation of **Sendai Framework for Disaster Risk Reduction**, this will reduce the vulnerability of any disaster.

(e) Focusing on Urban Flood Management : Keeping in view the fact that the problem of Urban Flooding is becoming more severe and losses are mounting every year.

- The subject of urban flooding needs exclusive attention and the proper implementation of **NDMA guidelines on Urban Flooding 2016,** is the need of the hour.

4.8.1 CAUSES OF FLOODS :

Primarily all floods are due to the surface runoff. Actually the floods are the results of a favourable combination of precipitation and the characteristics of the water shed.

The causes of flooding in all the major river systems are :

1. Intense rainfall
2. Topography of the catchment
3. Sedimentation of rivers and reservoirs
4. Obstruction in the river flow
5. Failure of dam
6. Failure of river embankment
7. Inadequate cross drainage works
8. Contraction of waterway

1. Intense rainfall :

An intense rainfall can occur only over a comparatively small area. Whenever such a precipitation bursts over a small watershed, it results in a high surface runoff, which reaches the main channel faster than it can be discharged, flooding the low lying area along the stream. The seriousness of the flood will depend upon the following factors :

- The intensity and duration of precipitation
- Surface slopes of the watershed
- Nature of the surface soil and covering vegetation

The intense precipitation sometimes also known as **cloud burst.**

If the rainfall is very heavy and the strom duration is longer, the resulting surface runoff will be more causing flood like situation.

2. Topography of the catchment :

For mountainous regions the relation of intensity of rainfall and slope of the surface have been found to have a good relationship. The same intensity of rainfall produces more discharge in a mountainous region than plain area.

The shape of the catchment also affect the surface runoff and flood. For a fan-shaped catchment, the time of concentration will be less and hence the peak flow will be more. But, in case of fern-leaf type (elongated) catchment of the same area and same storm, the time of concentration will be more. All the tributaries does not reach the stream at the same time, thus the peak flow will be comparatively less.

The more pervious the soil, the greater is its water retention power and longer the period of saturation, resulting in lesser floods.

3. Sedimentation of rivers and reservoirs :

If the top soil layer in the catchment area is loose and the vegetation cover is less, the tributaries flowing through such areas carry heavy sediment load. This sediment load is deposited in the river bed and

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he reservoir in which the river merges. Thus the cross-section of the river become shallow resulting in vertopping of water during heavy rains. Due to siltation in reservoirs, the storage capacity of the eservoirs decreases, resulting in flooding of the downstream area.

4. **Obstruction in the river flow :**

Sometimes, due to heavy landslides in the river, the natural flow of water is obstructed resulting in flood situation on the upstream of the river. When the obstructions in the river are removed due to heavy water pressure accumulated on the upstream, flooding is caused on the downstream of the river.

5. **Failure of dam :**

A dam is constructed to store huge quantity of water in the reservoir created on the upstream of the dam. Due to failure of the dam the large quanity of water stored in the reservoir instantly released causing flood in the downstream of the dam.

A dam may fail due to the following reasons :

- Piping
- Erosion
- Foundation failure
- Earthquake, etc.

The flood due to failure of the dam is called **flash flood.** This flash flood starts suddenly with tremendous velocity and depth, the disaster management or mitigation of which becomes virtually impossible.

For example, failure of Machhu dam of Morbi Gujarat in 1979.

A flash flood hydrograph and normal flood hydrograph are shown in Fig. 4.7.

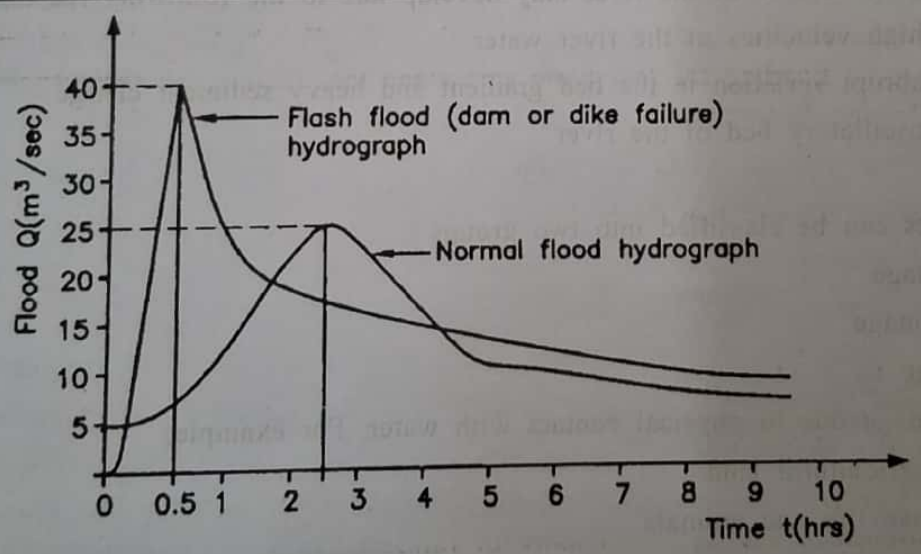

Fig. 4.7 Normal flood and flash flood bydrographs

6. **Failure of river embankment :**

Sometimes, the river embankment may fail due to gradual erosion by flowing water, resulting in flowing of flood water in the surrounding areas and flooding.

7. **Inadequate cross drainage works :**

Sometimes at the crossing of canal and river, it may be required to pass the river below the canal, by constructing a aqueduct. Due to the construction of an aqueduct, the depth of flow is reduced. Thus, during high flood, water level may rise on the u/s side and may submerge the surrounding area.

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8. **Contraction of waterway :**

Due to construction of railway or highway bridge across the river, the natural water way of river is contracted due to bridge piers and abutments. Thus, water rises on the u/s side due to insufficient passage and the upstream area may get flooded.

4.8.2 Ill-effects of floods :

The ill-effects of floods can be classified into the following three groups :

1. Inundation of low lying areas
2. Erosion of river banks
3. Change of river course

1. **Inundation of low lying areas :**

The following factors may cause the inundation of low lying areas :

(i) excessive runoff and inadequate channel capacity

(ii) Presence of obstructions in the stream

(iii) Inadequate waterway due to rail, road or canal crossing.

(iv) Congestion of confluences

(v) Tides resisting flood flows

2. **Erosion of river banks :**

The fast flowing current may eat away the banks, formed of easily erodable soil and thus may affect vast stretches of agricultural land.

3. **Change of river course :**

The change in the course of the river may develop due to the following reasons :

(i) Due to high velocities of the river water

(ii) Due to abrupt variation in the bed gradient and heavy sediment charge.

(iii) Due to oscillatory bed of the river

Flood Damages :

Flood damages can be classified into two groups :

1. Direct damage
2. Indirect damage

1. **Direct damage :**

It implies damage due to physical contact with water. For example,

- Erosion of agricultural land
- Loss of human life and animals
- Loss of property
- Breach of roads, canals, ponds, etc.
- Loss of forest and wild life

2. **Indirect damage :**

In this case flood water does not come in direct contact of the property, etc. but causes damage interruption and other causes.

For example,

- a labour is put out of employment or work due to submergence of the factory.
- Schools and colleges remains closed for few days due to flood.
- Due to interruption of transportation services, business industry is affected.

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4.8.3 Flood alleviation or flood mitigation or Flood Control :

A flood is a relatively high stage of river which is usually accompanied by overtopping of the natural banks of the river in a particular reach. The flood occurs because of heavy rains or melting of snow when the discharge in the river is so high that its natural cross-section is unable to contain it. Flood control is required to reduce the flood damage. Flood control is also known as flood management.

In India about 40 Mha of land is flood prone, which is about 12% of the total geographical area of 328 Mha.

There are two types of measures to mitigate the flood disaster :

1. Structural mitigation measures
2. Non-structural mitigation measures

Structural mitigation measures :

The structural mitigation measures are :

1. Storage reservoirs
2. Levees — low wall built at the side of a river to prevent it from flooding
3. Flood Walls
4. Channel improvement works
5. Diversion of flood water to flood ways
6. Construction of high earth platforms
7. Sluices

4.8.4 Non-structural mitigation measures : (June-2022)

The non-structural flood mitigation measures are:

1. Flood plain zoning
2. Flood forecasting and warning
3. Emergency evacuation of area
4. Management of flood plain
5. Flood proofing
6. Mathematical Modelling

1. Flood plain zoning :

Flood plain zoning means dividing the entire flood plain area into different zones and to restrict the occupancy of the different zones of the flood plain to uses which will suffer little or no damage during floods.

The objectives of flood plain zoning are :

(i) To restict encroachment of the zone which frequently comes under the influence of flood.

(ii) To direct selected activities in the zone which come under the influence of flood less frequently in such a manner that they are susceptible to flood damage.

(iii) To demarcate the zones which are subjected to flood only on rare occasions.

The flood zones are usually classified as follows :

- Prohibitive zone
- Restrictive zone
- Warning zone

2. Flood forecasting and warning :

Flood forecasting or warning, if could be conveyed earlier, disaster due to the probable flood can be

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minimized to a great extent. Temporary evacuation of persons and shifting the important property to safe places could be done before the flood arrives. Once the flood occurs, the normal activities of the society are badly disrupted with immense losses. Therefore, flood forecasting is a real necessity to minimize the flood havoc.

In India, seven flood forecasting units have been operating at

- New Delhi
- Guwahati
- Jalpaiguri
- Patna
- Bhubaneswar
- Surat
- Lucknow

There are another 22 substations, and they have been functioning to give timely warning of floods in some major rivers. For the forecasting the flood, historical data of the river like stream flow record, stage-discharge relationship, rainfall-runoff relationship are essential. A communication network is essential between the control room with gauge stations. Sometimes telephonic communicating lines fail during heavy storms. So wireless network become essential.

Forecasting of flood magnitude is evaluated by the following methods :

1. Rainfall-runoff relationship
2. Unit Hydrograph (UH)
3. Synthetic unit Hydrograph (SUH)
4. Flood routing model both hydrological and hydraulic
5. Stage discharge relationship

3. Emergency evacuation of area :

Sometimes emergency evacuation of the threatened area is one of the most effective means of reducing flood damage. This technique is adopted for sparsely populated areas where the property is also low valued. These areas do not justify other expensive methods of flood control. Loss of life can be prevented by emergency evacuation.

4. Management of flood plain :

The flood plain of a river is formed by sediment deposition or removal which usually occurs due to intermittent overflows of the stream above its banks. Such overflows usually occur every 2 or 3 years when the stream occupies some part of the flood plain. Because the flood plains are usually level surface they are attractive sites for roads and cities.

The various tools of flood plain management are as follows :

(i) Flood hazard surveys
(ii) Flood plain zoning
(iii) Flood proofing
(iv) Flood insurance

5. Flood proofing :

It essentially consists of a combination of both structural change and emergency action. Structural change includes construction of the building wall with some water proofing material, closure of lower level windows, construction of the house with the arrangement of stilts, i.e. an arrangement with bracket to raise the house when required. In some indigenous process, dwelling houses in flood areas, floor of the house is constructed at some upper level, even moderate flood is not able to touch the floor of the houses. During flood period people arrange their shelter in their own house. Impact of water pressure below the floor level is almost negligible as only the columns of the house are exposed.

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4.9 WATER RESOURCES CONSERVATION SYSTEM (Dec.-2021)

Water Resources of India

India accounts for about 2.45 per cent of the world's surface area, 4 per cent of the world's water resources and about 16 per cent of the world's population. The total water available from precipitation in the country in a year is about 4,000 cubic km. The availability from surface water and replenishable groundwater is 1,869 cubic km. Out of this, only 60 per cent can be put to beneficial uses. Thus, the total utilizable water resource in the country is only 1,122 cubic km.

The per capita availability of water is dwindling day-by-day due to increase in population. The available water resources are also getting polluted with industrial, agricultural and domestic effluents, and this, in turn, is further limiting the availability of usable water resources.

Water Conservation and Management -Need

Since there is a declining availability of freshwater and increasing demand, the need has arisen to conserve and effectively manage this precious life giving resource for sustainable development. Given that water availability from sea/ocean, due to high cost of desalinization, is considered negligible, India has to take quick steps and make effective policies and laws, and adopt effective measures for its conservation. Besides developing water-saving technologies and methods, attempts are also to be made to prevent the pollution. There is need to encourage watershed development, rainwater harvesting, water recycling and reuse, and conjunctive use of water for sustaining water supply in long run.

✓ **Methods for water resources conservation :** (June-2022)

(1) Prevention of Water Pollution

Available water resources are degrading rapidly. The major rivers of the country generally retain better water quality in less densely populated upper stretches in hilly areas. In plains, river water is used intensively for irrigation, drinking, domestic and industrial purposes. The drains carrying agricultural (fertilizers and insecticides), domestic (solid and liquid wastes), and industrial effluents join the rivers. The concentration of pollutants in rivers, especially remains very high during the summer season when flow of water is low.

The Central Pollution Control Board (CPCB) in collaboration with State Pollution Control Boards has been monitoring water quality of national aquatic resources at 507 stations. The data obtained from these stations show that organic and bacterial contamination continues to be the main source of pollution in rivers. The Yamuna river is the most polluted river in the country between Delhi and Etawah. Other severely polluted rivers are: the Sabarmati at Ahmedabad, the Gomti at Lucknow, the Kali, the Adyar, the Cooum (entire stretches), the Vaigai at Madurai and the Musi of Hyderabad and the Ganga at Kanpur and Varanasi. Groundwater pollution has occurred due to high concentrations of heavy/toxic metals, fluoride and nitrates at different parts of the country.

The legislative provisions such as the Water (Prevention and Control of Pollution) Act 1974, and Environment Protection Act 1986 have not been implemented effectively. The result is that in 1997, 251 polluting industries were located along the rivers and lakes. The Water Cess Act, 1977, meant to reduce pollution has also made marginal impacts. There is a strong need to generate public awareness about importance of water and impacts of water pollution. The public awareness and action can be very effective in reducing the pollutants from agricultural activities, domestic and industrial discharges.

(2) Recycle and Reuse of Water black water / grey water

Another way through which we can improve fresh water availability is by recycle and reuse. Use of water of lesser quality such as reclaimed wastewater would be an attractive option for industries for cooling and fire fighting to reduce their water cost. Similarly, in urban areas water after bathing and washing utensils can be used for gardening. Water used for washing vehicle can also be used for gardening. This would conserve better quality of water for drinking purposes. Currently, recycling of water is practiced on a limited scale. However, there is enormous scope for replenishing water through recycling.

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(3) Watershed Management

Watershed management basically refers to efficient management and conservation of surface and groundwater resources. It involves prevention of runoff and storage and recharge of groundwater through various methods like percolation tanks, recharge wells, etc. However, in broad sense watershed management includes conservation, regeneration and judicious use of all resources – natural (like land, water, plants and animals) and human with in a watershed. Watershed management aims at bringing about balance between natural resources on the one hand and society on the other. The success of watershed development largely depends upon community participation.

The Central and State Governments have initiated many watershed development and management programmes in the country. Some of these are being implemented by nongovernmental organisations also. Haryali is a watershed development project sponsored by the Central Government which aims at enabling the rural population to conserve water for drinking, irrigation, fisheries and afforestation. The Project is being executed by Gram Panchayats with people's participation.

Neeru-Meeru (Water and You) programme (in Andhra Pradesh) and Arvary Pani Sansad (in Alwar, Rajasthan) have taken up constructions of various water-harvesting structures such as percolation tanks, dug out ponds (Johad), check dams, etc., through people's participation. Tamil Nadu has made water harvesting structures in the houses compulsory. No building can be constructed without making structures for water harvesting.

Watershed development projects in some areas have been successful in rejuvenating environment and economy. However, there are only a few success stories. In majority of cases, the programme is still in its nascent stage. There is a need to generate awareness regarding benefits of watershed development and management among people in the country, and through this integrated water resource management approach water availability can be ensured on sustainable basis.

(4) Rainwater Harvesting

Rainwater harvesting is a method to capture and store rainwater for various uses. It is also used to recharge groundwater aquifers. It is a low cost and eco-friendly technique for preserving every drop of water by guiding the rain water to bore well, pits and wells. Rainwater harvesting increases water availability, checks the declining groundwater table, improves the quality of groundwater through dilution of contaminants, like fluoride and nitrates, prevents soil erosion, and flooding and arrests salt water intrusion in coastal areas if used to recharge aquifers.

Rainwater harvesting has been practiced through various methods by different communities in the country for a long time. Traditional rainwater harvesting in rural areas is done by using surface storage bodies, like lakes, ponds, irrigation tanks, etc. In Rajasthan, rainwater harvesting structures locally known as Kund or Tanka (a covered underground tank) are constructed near or in the house or village to store harvested rainwater.

There is a wide scope to use rainwater harvesting technique to conserve precious water resource. It can be done by harvesting rainwater on rooftops and open spaces. Harvesting rainwater also decreases the community dependence on groundwater for domestic use. Besides bridging the demand supply gap, it can also save energy to pump groundwater as recharge leads to rise in groundwater table. These days rainwater harvesting is being taken up on massive scale in many states in the country. Urban areas can specially benefit from rainwater harvesting as water demand has already outstripped supply in most of the cities and towns.

(5) Redistribution of Water (River linking):

Water found on the surface of the earth is not equally distributed. Existing form of distribution also becomes a reason for the water crisis.

There is maximum rainfall of the world (1,187 cm. in Mawsynram) in north-east India, whereas there is only 50 cm. rainfall in the west. As a result of it, more than 60 per cent portion of water of Brahmaputra

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and its companion rivers in the north east flows to the saline seas uselessly, whereas rivers of western Rajasthan remain dry for most of the time in a year.

Hence, by arranging supply of water from areas having lesser demand to the areas having greater demand, water crisis can be minimized. By construction of surface water reservoirs and storage of excess water in them, supply can be made to scarcity affected areas. This work can be accomplished by development of water reservoirs and canal network.

Excess rainfall water which flows away from rivers without being used, can be stored by construction of water reservoirs, from where it can be supplied for agriculture, industries, urban areas etc. Facilities of fisheries and transport also exist in stored water. Redistribution of water is also possible through canal system. Canal system transfers water from excess rain water areas to scarce rain water areas and conserves water for different uses. ***Indira Gandhi Canal*** is such a type of canal system which has brought water of Himalayas to western Rajasthan and changed the arid ecology.

(6) Rational use of ground water

Groundwater meets 25 per cent of total supply of water in the world, remaining 75 per cent supply is met by surface water sources of rivers, lakes etc. Demand for groundwater goes on increasing in proportion to its available quantity due to which quantity of groundwater goes on decreasing. After exploitation of groundwater, its re-infiltration takes a very long time to complete.

Hence, groundwater exploitation should be only in proportion to its recharging capacity. In India, the maximum groundwater is utilized for agriculture. Instead of growing crops requiring water according to agricultural climatic conditions, crops of commercial importance requiring more water are grown, due to which overexploitation of groundwater is done.

By making only conscientious exploitation of groundwater, it can be proportionately conserved. For that, crops requiring lesser water should be given priority over crops requiring more water. Lesser water should be used by industries. For recharging ground-water, necessary drainage area should also be made available.

(7) Renovation of Traditional Water Sources:

In India, traditional water storage places have been able to meet the demand of drinking water in many regions but they have been renovated from time to time. Water stored in traditional water sources has been used for both purposes, agriculture as well as for drinking.

Important traditional water conservation methods for irrigation include Kuhul in hilly areas, Jing (Ladakh), Kool, Water Kundis called Khoop in Arunachal Pradesh, Zabo method of Nagaland, Aabi tanks of Haryana, Dong Pokhar of Assam, Bandhare of Maharashtra, Kere of Karnataka, Iree (Tanks) in Tamil Nadu, jackwell of Andaman Nicobar and Nadi, Tanka, Kund, Khadeen, Kui, Bedi, Baori, Jhalra, Toba etc. of Rajasthan.

(8) Use of Modern Irrigation Methods:

At the world level, 69 per cent water is used for the agricultural sector. Necessary requirement of water in agriculture is met by surface water sources and groundwater. Surface water used for agriculture is obtained through canals and tanks and underground water from wells and tube wells. Unfortunately, much water is wasted through traditional methods of irrigation. Canal irrigation does not distribute water in a balanced way and it creates problems of water logging.

A large portion of water can be conserved by adoption of modern methods of irrigation. Irrigation consumes double the quantity of water in comparison to all other uses. Sprinkler and drip irrigation methods save 50 per cent water.

(9) Increasing forest cover

Due to increasing deforestation during the last century, most of the rainwater flowed away to the saline seas without infiltrating into the ground. Water crisis also developed during the last decade in Cherapunji

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which gets highest rainfall in the world, because forest cover has been destroyed there due to mining of limestone. As a result of it, rain water flows away very fast to the rivers. A similar thing is happening in the Dehradun area of Uttaranchal.

The old tradition of tree plantation on the banks of rivers and tanks will have to be revived. Forest cover will have to be developed on uncultivable waste lands and hilly slopes on a large scale. Since trees bear drought conditions for a long duration as compared to crops, hence trees are helpful in reducing the demand for water along with recharging water sources.

(10) Water Conservation by Municipal Bodies:

Municipal bodies should manage both individual demand and supply of water as well as conserve water. Municipal laws should provide for collection of rain water from roof tops and implement it. Individual awareness is very important in water conservation.

Every individual should develop the attitude of water conservation and prevent every drop of water from being wasted. As far as possible, one should conserve rain water along with maintaining quality of the water. Depth of traditional water sources should be maintained by controlling its cleanliness.

(11) By Individuals

A little water conservation methods that can be practiced by individuals to reduce the wastage of water are provided below.

- Installing flow-restricting shower heads to save water during showers.
- Taking bucket-baths instead of showers.
- Turning off the tap while shaving or brushing teeth.
- Immediately fixing any leaking taps and pipes in our homes.
- Practicing rainwater harvesting to reduce the wastage of rainwater.

REVIEW QUESTIONS

1. What do you mean by smart water management ? What are its goals and tools ? **(June-2022)**
2. Describe various smart water management technologies.
3. What are the benefits associated to water and wastewater management ?
4. Write short note on: Water smart city(WSC).
5. Write short note on: Rainwater harvesting.
6. Describe structures for the storage of water.
7. Write short note on: Conveyance of water. **(Dec.-2021)**
8. Describe sustainable water and sanitation. What are issues and challenges of urban sanitation. **(Dec.-2021, June-2022)**
9. Describe Innovative Sanitation Solutions for smart cities.
10. Discuss waste water generation scenario in India and need for wastewater treatment.
11. Describe IoT based smart wastewater management systems for smart cities. **(Dec.-2021, June-2022)**
12. Write short note on: Integrated flood management system.
13. What are the causes of floods and what are its effects.
14. Describe various flood mitigation measures. **(June-2022)**
15. Describe flood forecasting and warning in India. **(Dec.-2021, June-2022)**
16. Describe various methods for water resources conservation. **(Dec.-2021, June-2022)**
17. Write a note on water-meters. **(Dec.-2021)**
18. Give your suggestions to improve existing water conveyance system. **(Dec.-2021, June-2022)**

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